

Does ICU-level IMV quality explain between-unit variation in ventilated-patient outcomes?

Project concept – CLIF Consortium – draft

Aim

Quantify between-ICU variation in risk-adjusted survival and ventilator-free days (VFD) among invasively ventilated patients, and estimate the fraction of that variation attributable to a unit-level summary of invasive mechanical ventilation (IMV) quality, the CLIF_quality_score.

Setup and notation

- **Cohort:** all adult patients j in ICU unit i ($i = 1, \dots, K$) receiving IMV for > 24 h.
- **Exposure:** Q_i , the CLIF_quality_score of unit i (standardized, mean 0, SD 1).
- **Adjustment:** $\mathbf{X}_{ij}$, the six SOFA components (respiratory, coagulation, liver, cardiovascular, CNS, renal), worst values in the first 24 h.
- **Outcomes:** $Y_{ij} \in \{0, 1\}$, survival (hospital discharge alive); $V_{ij} \in [0, 28]$, VFD at day 28 (death $\Rightarrow V_{ij} = 0$).

Model 1: unit random intercept, case-mix adjustment only

$$\text{logitPr}(Y_{ij} = 1 \mid u_i, \mathbf{X}_{ij}) = \beta_0 + u_i + \beta^\top \mathbf{X}_{ij}, \quad u_i \sim \mathcal{N}(0, \sigma_u^2) \quad (1)$$

$$V_{ij} = \gamma_0 + v_i + \gamma^\top \mathbf{X}_{ij} + \varepsilon_{ij}, \quad v_i \sim \mathcal{N}(0, \sigma_v^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2) \quad (2)$$

Unit-specific predictions for a reference (mean) patient $\bar{\mathbf{X}}$, using empirical Bayes estimates $\hat{u}_i, \hat{v}_i$:

$$\hat{p}_i = \text{expit}(\hat{\beta}_0 + \hat{u}_i + \hat{\beta}^\top \bar{\mathbf{X}}), \quad \hat{\mathbb{E}}[V \mid i, \bar{\mathbf{X}}] = \hat{\gamma}_0 + \hat{v}_i + \hat{\gamma}^\top \bar{\mathbf{X}} \quad (3)$$

Figure 1 (caterpillar plot): units ranked by $\hat{p}_i$ (and separately by $\hat{\mathbb{E}}[V \mid i]$) with 95% prediction intervals, points colored by Q_i . Visual test: are high- Q units concentrated at the favorable end?

Model 2: add unit-level IMV quality

$$\text{logitPr}(Y_{ij} = 1 \mid u_i, \mathbf{X}_{ij}, Q_i) = \beta_0 + u_i^* + \beta_1 Q_i + \beta^\top \mathbf{X}_{ij}, \quad u_i^* \sim \mathcal{N}(0, \sigma_{u^*}^2) \quad (4)$$

$$V_{ij} = \gamma_0 + v_i^* + \gamma_1 Q_i + \gamma^\top \mathbf{X}_{ij} + \varepsilon_{ij}, \quad v_i^* \sim \mathcal{N}(0, \sigma_{v^*}^2) \quad (5)$$

Proportional change in between-unit variance, and median odds ratio (MOR) for survival:

$$\text{PCV}_Y = \frac{\hat{\sigma}_u^2 - \hat{\sigma}_{u^*}^2}{\hat{\sigma}_u^2}, \quad \text{PCV}_V = \frac{\hat{\sigma}_v^2 - \hat{\sigma}_{v^*}^2}{\hat{\sigma}_v^2}, \quad \text{MOR} = \exp\left(\sqrt{2\sigma^2} \Phi^{-1}(0.75)\right) \quad (6)$$

Hypotheses

H1: $\beta_1 > 0$ and $\gamma_1 > 0$ (higher unit IMV quality $\Rightarrow$ higher odds of survival, more VFD). **H2:** PCV_Y and PCV_V are large, i.e. Q_i explains a substantial share of between-unit variation.

Key threats to validity (to resolve before analysis)

- **Mechanical coupling:** if Q_i is computed from the same patients, components linked to liberation (e.g. SBT performance) partly are VFD. Compute Q_i leave-one-patient-out or from a prior time period.
- **Cohort selection on a post-exposure variable:** “IMV > 24 h” excludes early extubations and early deaths, both plausibly caused by unit quality. Report sensitivity with IMV ≥ 0 h and landmarking.
- **Post-treatment adjustment:** respiratory SOFA (P/F) and cardiovascular SOFA depend on unit practice (PEEP, F_iO_2 , vaso-pressor thresholds); adjustment may absorb part of the effect.
- **Unit-level confounding:** Q_i may proxy for volume, staffing, academic status, or data completeness across CLIF sites; PCV is not a causal decomposition. Add unit/hospital covariates and a hospital level (units nested in hospitals).
- **VFD is not Gaussian:** bounded, zero-inflated by death. Prefer a competing-risks (liberation vs. death) or ordinal/hurdle model; treat the linear model as descriptive.
- **“Mean patient” $\neq$ average patient:** under a logit link, $\text{expit}(\cdot \bar{\mathbf{X}})$ differs from the marginal risk; report standardized (population-averaged) unit estimates as well.